

AI, algorithmic decision making and Big Data: Risks and Opportunities

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The Internet, AI and Big Data: promise and catch

The Internet & AI infrastructure can deliver good:

- It improves efficiency and effectiveness in many domains (smart cities, e-health, etc.)
- It allows for a world-wide generation and distribution of knowledge and solutions
- We can discover new correlations between things:
 - Doctors can provide better diagnoses and personalised and targeted therapies
- Cost savings, greater productivity, and value creation:
 - Firms can anticipate market trends and make more efficient decisions
 - Consumers can make better informed choices and obtain personalised services

AI opportunities: techno-optimistic perspective

Technologies based on artificial intelligence can allow humans to face the ***“the grand challenges of humanity, such as maintaining a healthy environment, providing the resources for a growing population (including energy, food, and water), overcoming disease, vastly extending human longevity, and eliminating poverty”***. (Ray Kurzweil, *How to Create a Mind*)



AI opportunities: techno-optimistic perspective

- AI can fully replace human activities (e.g., in driverless vehicles, cleaning robots, and certain planning and scheduling tasks in logistics).
- In most cases, however, it rather complements human capacities: it enhances the human ability to know and act, it supports creativity and invention.
- Thanks to AI, it becomes possible to achieve between humans and machines a new cooperation that overcomes the classic model on which machines only performed routine and repetitive tasks.



AI and Big Data risks

- Eliminate or devalue the jobs of those who can be replaced by machines (exclusion and marginalization in the job market)
- Lead to poverty and social exclusion (e.g., AVs and taxi drivers, chatbot and call center workers)
- Favour economic models in which “*the winner takes all*”.
 - *Huge profits+limited workforce => AI contributes to concentrating wealth in those who invest in such companies or provide them with high-level expertise.*
- New opportunities for illegal activities
 - In particular, AI & Big Data systems can fall subject to **cyberattacks** (designed to disable critical infrastructure, or steal or rig vast data sets, etc.), and they can even be **used to commit crimes** (e.g., autonomous vehicles can be used for killing or terrorist attacks, and intelligent algorithms can be used for fraud or other financial crimes).

AI and Big Data risks

➤ Pervasive surveillance and manipulation

- To satisfy data-hungry AI applications, the Internet has become an infrastructure for data collection (and surveillance)
- All facts, even the apparently insignificant ones, are useful for learning algorithms, scalability is no problem

The power of AI can be used to pursue **economic interests in ways that are harmful to individuals and society**: users, consumers, and workers can be subject to pervasive control, limitations in their access to information and opportunities, manipulation in their choices.

Certain abuses may be incentivised by the fact that many tech companies—such as major platforms hosting user-generated content— operate in **two- or many-sided markets**.

AI and Big Data risks

- ❖ Their main services (search, social network management, access to content, etc.) are offered to individual consumers, but the revenue stream comes from advertisers, influencers, and opinion-makers (e.g., in political campaigns).
- ❖ This means not only that any information that is useful for targeted advertising will be collected and used for this purpose, but also that platforms will employ any means to **capture users**, so that they can be exposed to ads and attempts at persuasion.
- ❖ This may lead not only to a massive collection of personal data about individuals, to the detriment of privacy, but also to a **pervasive influence on their behavior, to the detriment of both individual autonomy and collective interests**.

AI and Big Data risks

- Polarization and fragmentation in the public sphere
 - proliferation of sensational and **fake news**, when used to capture users by exposing them to information they may like, or which accords with their preferences, thereby exploiting their confirmation biases .
- Just as AI can be misused by economic actors, it can also be **misused by the public section**. Governments have many opportunities to use AI for legitimate political and administrative purposes (e.g., efficiency, cost savings, improved services), but they may also **employ it to anticipate and control citizens' behaviour in ways that:**
 - Restrict individual liberties and interfere with the democratic process
 - Unfairness, discrimination and inequality

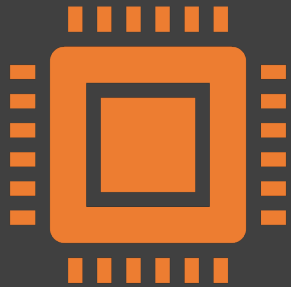


AI, Influence and Manipulation

Profiling, influence and manipulation

- The use of automated assessment systems may be problematic also where their performance is not worse, or even is better, than what humans would do.
- This is due to the fact that automation **diminishes the costs of collecting** information on individuals, **storing** this information and **process** it in order to **evaluate individuals and make choices accordingly**.
- Thus, automation paves the way for much more **persistent and pervasive mechanisms for assessment and control**.
- In general, thanks to AI, all kind of personal data can be used to **analyse, forecast and influence** human behaviour, an opportunity that transforms them into **valuable commodities**. Information that was not collected or was discarded as worthless “data exhaust” —e.g., trails of online activities—has now become a prized resource.

Profiling



Through AI & Big Data technologies—in combination with the panoply of sensor that increasingly trace any human activity—individuals can be subject to surveillance and influence in **many more cases and contexts**, on the basis of a broader set of personal characteristics (ranging from economic conditions to health situation, place of residence, personal life choices and events, online and offline behaviour, etc.).

By correlating data about individuals to corresponding classifications and predictions, AI increases the potential for **profiling**, namely, for inferring information about individuals or groups, and adopting assessments and decisions on that basis.

Profiling: the scenario

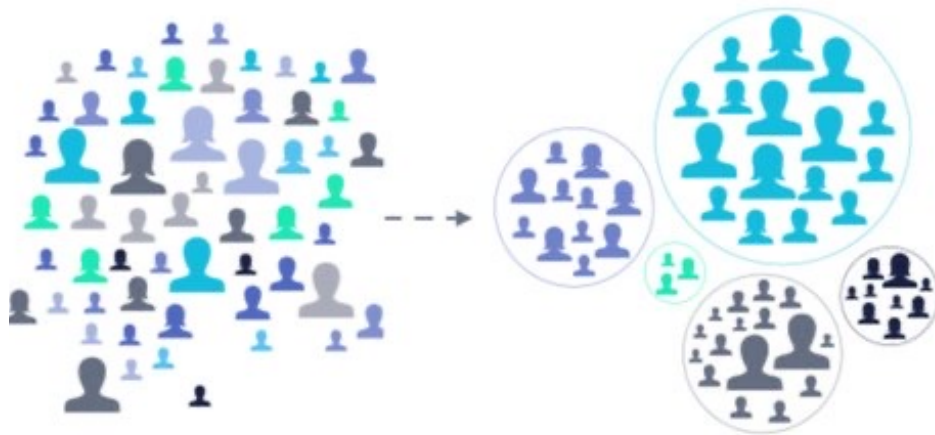


A profiling system establishes (predicts) that individuals having certain features F_1 , also have a certain likelihood of possessing certain additional features F_2 .

- For instance, assume that the system establishes (predicts) that those having a genetic patterns have the tendency to develop a higher than average chance to develop cancer, or that those having a certain education and job history or ethnicity have a certain higher-than-average likelihood to default of their debts). Then we may say that this system has profiled the group of the individuals possessing features F_1 : it has added to the description (the profile) of these group a new segment, namely, the likelihood of possessing the additional features F_2 .

Profiling: the scenario

Segmentation and Profiling



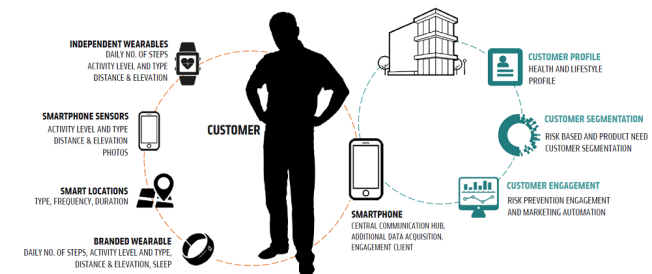
If the system is then given the information that a specific individual has features F_1 , then It can infer that it is likely that this individual also has feature F_2 . This may lead to the individual being treated accordingly, in a beneficial or a detrimental way.

- For instance, in the case in which the inferred feature of an individual is his or her **higher susceptibility to cancer**, the system's indication may provide the basis for **preventive therapies** and tests, or rather for a **raise in the insurance premium**.

AI and profiling

- **AI & Big Data have vastly increased the opportunities for profiling.**
- Through the training, the system has learned an algorithmic model that can be applied to new cases: **if the model is given predictors-values concerning a new individual, it infers a corresponding target value for that individual, i.e., a new data item concerning him or her.**
 - the creditworthiness of loan applicants on the basis of their financial history, their online activity and social condition;
 - the likelihood that convicted persons may reoffend on the basis their criminal history, their character (as identified by personality test) and personal background.

These predictions may trigger automated determinations concerning, respectively, the price of a health insurance, the granting of a loan, or the release on parole.





Profiling: influence and manipulation

- **The information so inferred may also be conditional**, that is, it may consist in the **propensity to react in a certain way to given inputs**.
 - For instance, it may consist in the propensity to respond to a therapy with improved medical condition, or in the propensity to respond to certain ads or to a certain price variation with a certain purchasing behaviour, or in the propensity to respond a certain kind of message with a change in mood or preference (e.g., relatively to political choices).
- **When that is the case, profiling potentially leads to influence and manipulation.**

Profiling: influence and manipulation



- Assume, too, that the system connects certain values for input features (e.g., having a certain age, gender, social status, personality type, etc.) to the propensity to react to a certain message (e.g., a targeted ad) with a certain response (e.g., buying a certain product). Assume also that the system is told that a particular individual has these values (he is a young male, working class, extrovert, etc.).
- Then the system would know that by administering to the individual that message, the individual can probably be induced to deliver the response.

Even when an automated assessment and decision-making system —a profile-based system— **is unbiased, and meant to serve beneficial purposes, it may negatively affect the individuals concerned.** Those who are subject to pervasive surveillance, persistent assessments and insistent influence come under heavy psychological pressure that affects their personal autonomy, and they are susceptible to deception, manipulation and exploitation in multiple ways.



Profiling: a notion

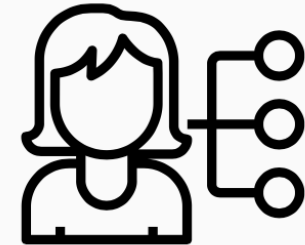
- Profiling is a technique of (partly) automated **processing of personal and/or non-personal data**, aimed at **producing knowledge by inferring correlations** from data in the form of profiles that can subsequently be **applied as a basis for decision-making**.
- A profile is a set of **correlated data** that represents a (individual or collective) subject.
- Constructing profiles is the process of **discovering unknown patterns** between data in large data sets that can be used to create profiles.
- Applying profiles is the process of **identifying and representing a specific individual or group as fitting a profile** and of taking some form of **decision** based on this identification and representation.

Bosco et al (2015); see also Hildebrandt, M. (2009).

Profiling in GDPR

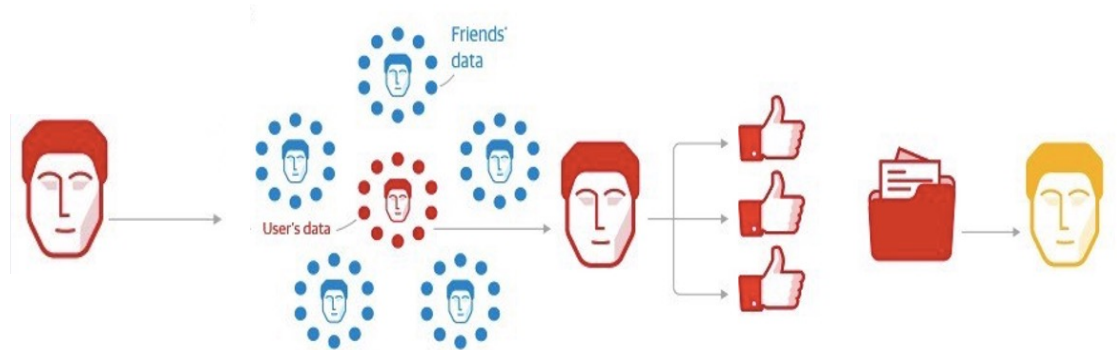
The notion of profiling in the GDPR only covers assessments or decisions concerning individuals, based on personal data, excluding the mere construction of group profiles:

- *'profiling'[...] consists of any form of automated processing of personal data evaluating the personal aspects relating to a natural person, in particular to analyse or predict aspects concerning the data subject's performance at work, economic situation, health, personal preferences or interests, reliability or behaviour, location or movements, where it produces **legal effects** concerning him or her or **similarly significantly affects him or her**.*



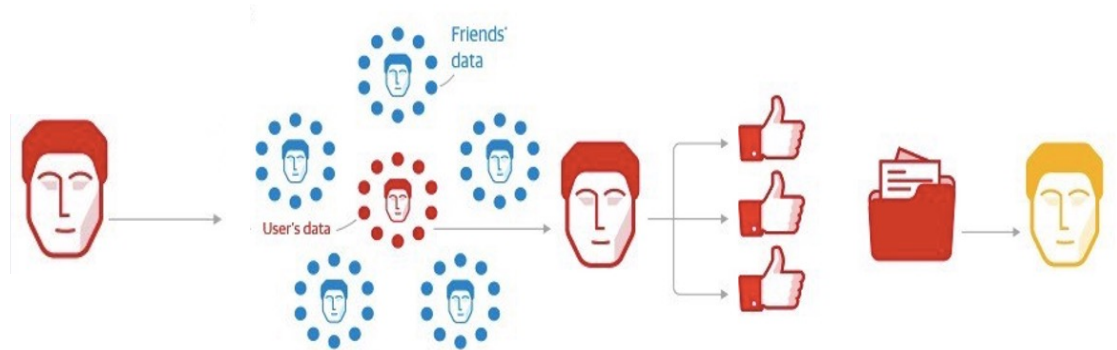
The dangers of profiling: the case of Cambridge Analytica

- First of all, people being registered as **voters** in the USA were invited to take a **detailed personality/political test** (about 120 questions), available online. The individuals taking the test would be **rewarded with a small amount of money** (from two to five dollars). They were told that their data would only be used for the academic research. About **320 000 voters took the test**. In order to receive the reward each individual taking the test had to **provide access to his or her Facebook page (step 1)**. This allowed the system to connect each individual's answers to the information included in his or her Facebook page.



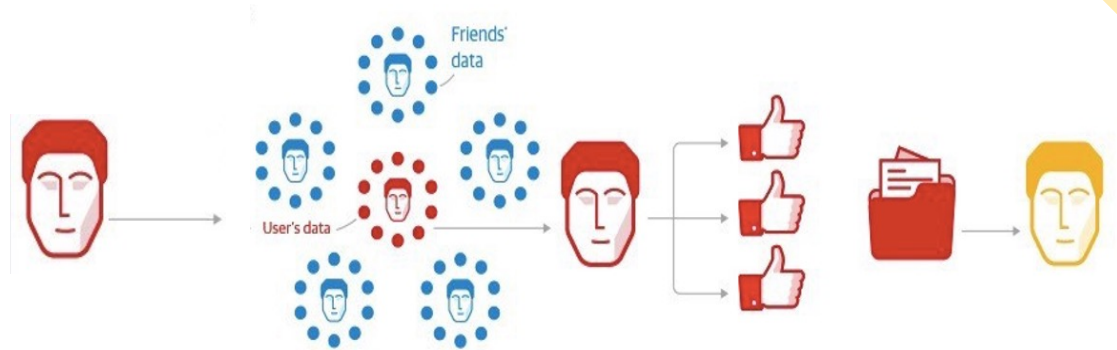
The dangers of profiling: the case of Cambridge Analytica

- When accessing a test taker's page, Cambridge Analytica **collected** not only **the Facebook page of test takers**, but also the Facebook **pages of their friends**, between 30 and 50 million people altogether (**step 2**). Facebook data was also collected from other sources.
- After this data collection phase, Cambridge Analytica had at its disposition **two sets of personal data** to be processed (**step 3**): (1) the data **about the test takers**, consisting in the information on their **Facebook pages**, **paired with their answers** to the questionnaire, (2) and the **data about their friends**, consisting only in the information on their **Facebook pages**.



The dangers of profiling: the case of Cambridge Analytica

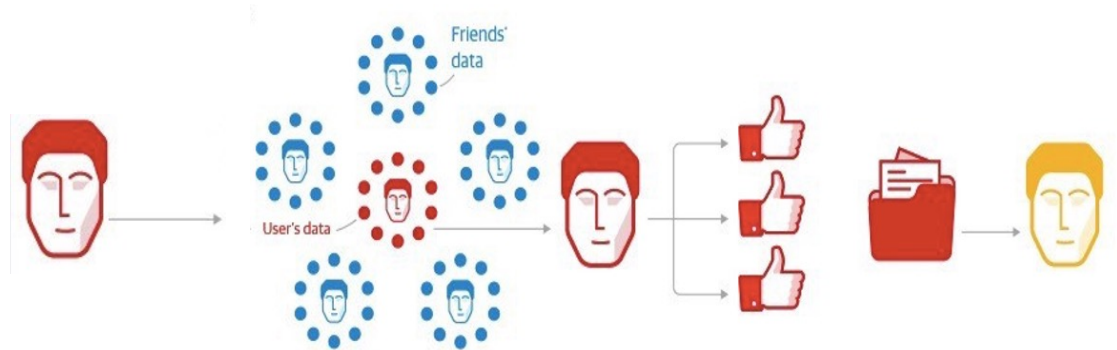
- Cambridge Analytica used the data about test-takers as a training set for building a model to profile their friends and other people. Data about the test-takers constituted a vast training set, where the information on an individual's **Facebook pages** (likes, posts, links, etc.) provided **values for predictors (features)** and **the answers to the questionnaire** (and psychological and political attitudes expressed by such answers) provided **values for the targets**. Thanks to its machine learning algorithms Cambridge Analytica could use this data to build a model **correlating the information in people's Facebook pages to predictions about psychology and political preferences**.



- Cambridge Analytica engaged in **massive profiling**, namely, in **expanding the data available on the people who did not take the test** (their Facebook data, and any further data that was available on them), **with the predictions provided by the model**. E.g. if test-takers having a certain pattern of Facebook likes and posts were classified as having a neurotic personality, the same assessment could be extended also to non-test-takers having similar patterns in their Facebook data.

The dangers of profiling: the case of Cambridge Analytica

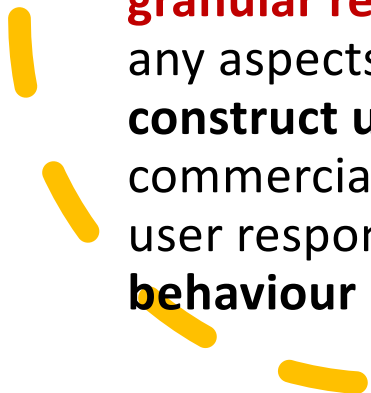
Finally (**stage 4**), based on this personality/political profiling, **potential voters who were likely to change their voting behaviour were identified** (initially **2m people in 11 US States** in which a small change could make a difference) if provided with appropriate messages. These **voters were targeted with personalised political ads** and with other messages that could trigger the desired change in voting behaviour, possibly building upon their emotions and prejudice and without making them aware of the purpose of such messages.





Towards surveillance capitalism or surveillance State?

- Some authors have taken a positive view of the development of systems based on the massive collection of information. They have observed that the integration of AI and Big Data enables **increased efficiency** and provides **new means for managing and controlling individual and social behaviour**.
- When economic transactions —and more generally social interaction and individual activities— are **computer-mediated**, they provide for a **ubiquitous and granular recording of data**: computer systems can observe, verify and analyse any aspects of the activities in question. **The recorded data can be used to construct user profiles, to personalise interactions with users** (as in targeted commercial communication), **to engage in experimentation** (e.g., to evaluate user responses to changes in prices and messaging), **to guide and control behaviour** (e.g., for the purpose of economic or political persuasion).





Towards surveillance capitalism or surveillance State?

In this context, **new models of economic and social interaction become possible, which are based on the possibility of observing every behaviour, and of automatically linking penalties and rewards to it.**

- Consider for instance how online consumers trust vendors of goods and services with whom they have never had any personal contact, relying on the platform through which such goods and services are provided, and on the platform's methods for **rating, scoring, selecting, and excluding.**

“A fascinating look at a new field by one of
its principal geeks.” —*The Economist*



SOCIAL PHYSICS



HOW SOCIAL NETWORKS CAN
MAKE US SMARTER

ALEX PENTLAND

Towards surveillance capitalism or surveillance State?

According to Alex Pentland the director of the Human Dynamics Lab at the MIT Media Lab, AI & Big Data may enable the development of a **“social physics”**, i.e., a rigorous social science. The availability of vast masses of data and of methods and computational resources to process these data could support a **social science having solid theoretical-mathematical foundations as well as operational capacities for social governance.**

Industrial Capitalism and Surveillance capitalism

The prospect for economic and social improvement offered by AI and Big Data is accompanied by the risks referred to as “**surveillance capitalism**” and the “**surveillance State**”.

According to **Shoshana Zuboff**, **surveillance capitalism** is the **leading economic model** of the present age.

Industrial Capitalism

Karl Polanyi observed that industrial capitalism also treats as **commodities** (products to be sold in the market) **entities that are not produced for the market**: **human life** becomes “**labour**” to be bought and sold, **nature** becomes “**land**” or “**real estate**”, **exchange** becomes “**money**.”

As a consequence, the dynamics of capitalism produces **destructive tensions —exploitation, destruction of environment, financial crises—** unless **countervailing forces**, such as **law, politics** and **social organisations** (e.g., workers’ and consumers’ movements), intervene to counteract, moderate and mitigate excesses.

Industrial Capitalism and Surveillance capitalism

The prospect for economic and social improvement offered by AI and Big Data is accompanied by the risks referred to as “**surveillance capitalism**” and the “**surveillance State**”.

According to **Shoshana Zuboff**, **surveillance capitalism** is the **leading economic model** of the present age.

Surveillance Capitalism

It expands commodification, extending it to **human experience**, which it turns into **recorded and analysed behaviour**, i.e., it transforms into **marketable opportunities to anticipate and influence**.

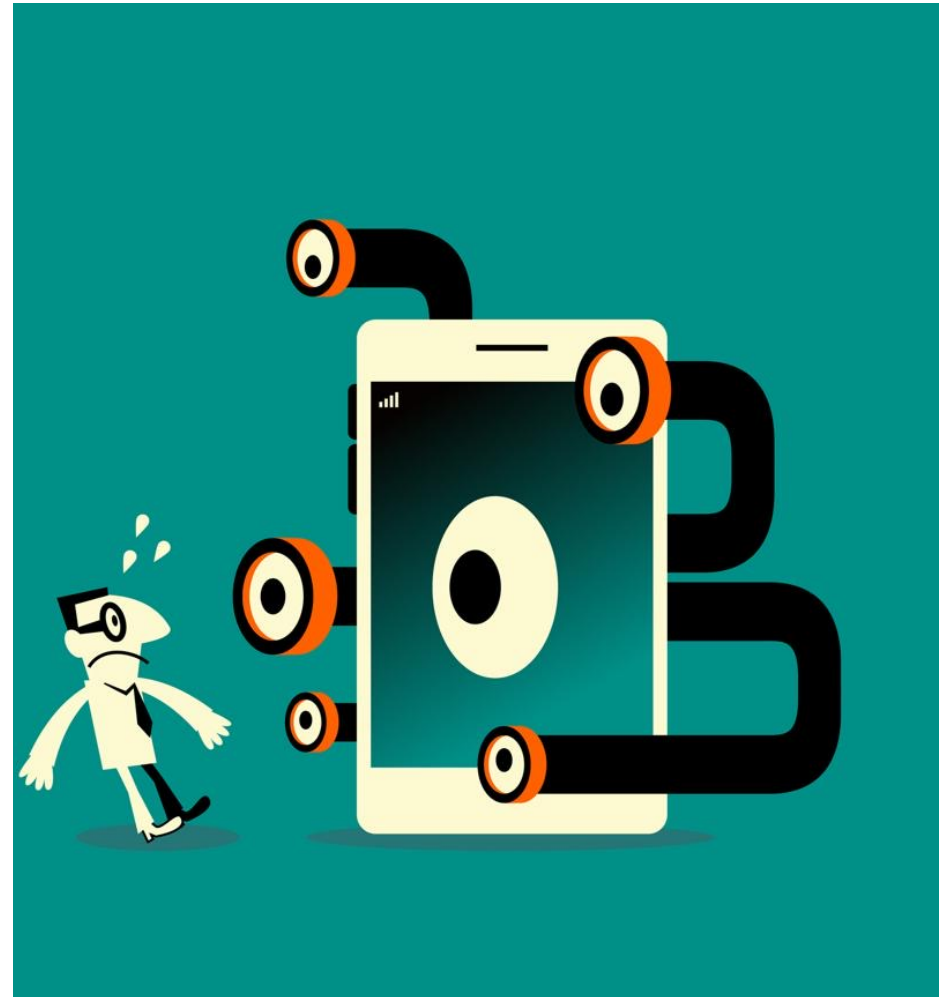
annexes **human experience** to the market dynamic so that it is reborn as behavior: **the fourth “fictional commodity.”**

Surveillance capitalism

Polanyi's first three **fictional commodities**—land, labor, and money—were subjected to **law**.

Although these laws have been imperfect, the institutions of labor law, environmental law, and banking law are regulatory frameworks intended to defend society (and nature, life, and exchange) from the worst excesses of raw capitalism's destructive power.

Surveillance capitalism's expropriation of human experience has faced **no** such **impediments**.





Surveillance capitalism

- In the case of surveillance capitalism, raw market dynamics can lead to novel disruptive outcomes. Individuals are subject to **manipulation**, are **deprived of control over their future** and **cannot develop their individuality**. Social networks for collaboration are replaced by surveillance-based mechanism of incentives and disincentives. (e.g. Uber recording workers' performance + mutual reviews of workers and clients)
- This new way of governing human behaviour may lead to efficient outcomes, but **it affects the mental wellbeing and autonomy of the individuals concerned**.
- According to Zuboff, we have **not yet** developed **adequate legal, political or social measures** by which to check the potentially disruptive outcomes of surveillance capitalism and keep them in balance. However, she observes, **the GDPR** could be an important step in this direction
- The need to **limit the commercial use of personal data** has led to new legal schemes not only in Europe, but also in California, the place where many world-leading "surveillance capitalists" have their roots; the CCPA (**California Consumer Privacy Act**), which came into effect on January 2020, provides consumers with rights to access their data and to prohibit data sales (broadly understood).

Surveillance State

- At the governmental level, surveillance capitalism finds its parallel in the so-called “surveillance State”
- In the National Surveillance State, the government uses surveillance, data collection, collation, and analysis to **identify problems**, to **head off potential threats**, to **govern populations**, and to **deliver valuable social services**.
- The National Surveillance State is a special case of **the Information State**-a state that tries to identify and solve problems of governance through the collection, collation, analysis, and production of information.



Surveillance State

Advantage



- Support efficiency in managing public activities
- Coordinate citizens' behaviour
- Prevent social harms

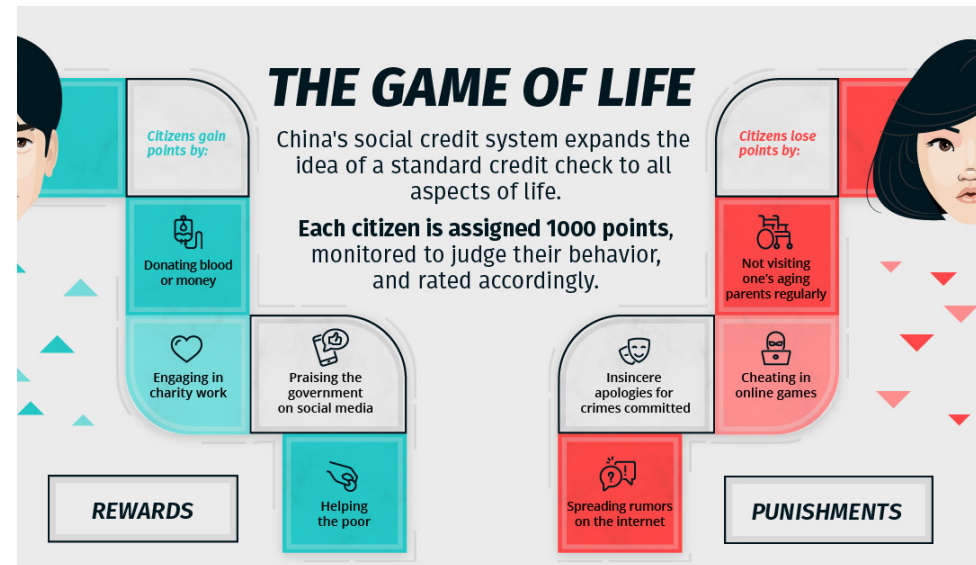
Disadvantage



- New kinds of influence and control
- Promote values and Purposes that may conflict with democracy
- Diminish autonomy

Surveillance State: the Chinese Social credit systems

- The Chinese Social credit systems collects data about citizens and assigns to those citizens scores that quantify their social value and reputation.
- It is based on the aggregation and analysis of personal information
- The collected data cover **financial aspects** (e.g., timely compliance with contractual obligations), **political engagement** (e.g., participation in political movements and demonstrations), **involvement in civil and criminal proceedings** (past and present) and **social action** (e.g. participation in social networks, interpersonal relationships, etc.).
- Citizens may be assigned positive or negative points, which contribute to their social score.



- The overall score determines citizens' **access to services and social opportunities**, e.g. universities, housing, transportation, jobs, financing, etc.
- The system's purported **objective** is to **promote mutual trust, and civic virtues**.
- **Risks: opportunism and conformism** may be rather promoted to the detriment of individual autonomy and genuine moral and social motivations.



Individual and social costs of AI & Big Data applications

In some cases and domain, AI & Big Data applications—even when accurate and unbiased—may have individual and social costs that outweigh their advantages.

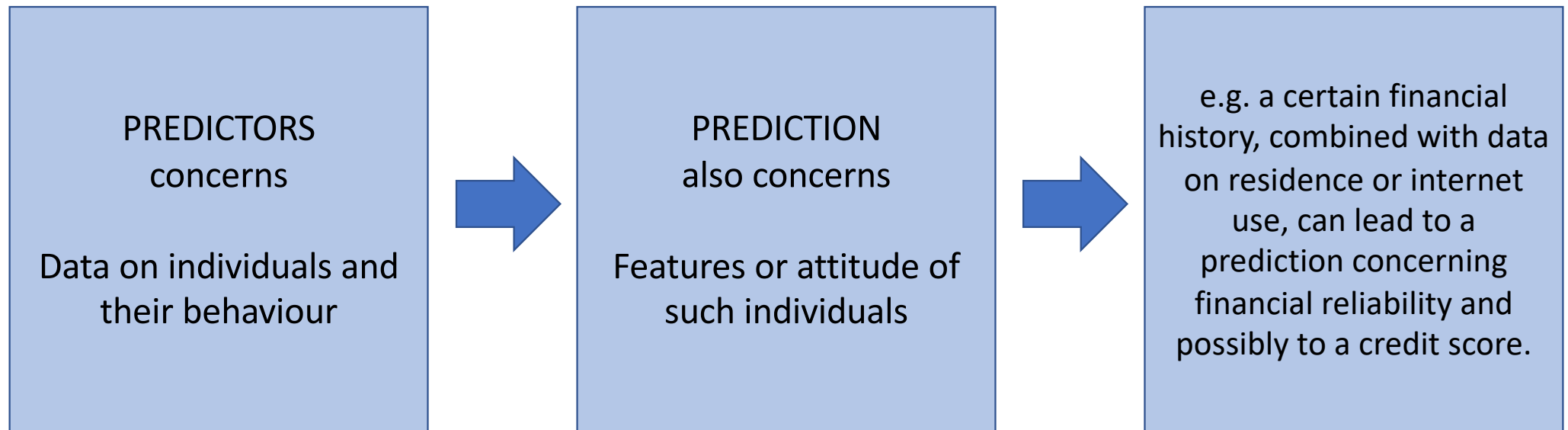
- Which systems really deserve to be built?
- Which problems most need to be tackled?
- Who is best placed to build them?
- And who decides?

We need genuine **accountability mechanisms**

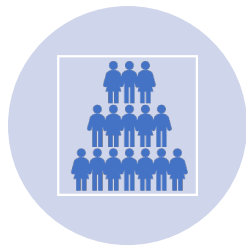
Consider, for instance, systems that are able to recognise sexual orientation, or criminal tendencies from the faces of persons. **Should we just ask whether these systems provide reliable assessments, or should we rather ask whether they should be built at all?**

The general problem of social sorting and differential treatment

The key aspect of ML systems is the ability to engage in **differential inference: different combinations of predictor-values are correlated to different predictions.**



The general problem of social sorting and differential treatment



A new dynamic of **stereotyping** and **differentiation** takes place.

(a) The individuals whose data support the same prediction, will be considered and treated in the same way.

(b) The individuals whose data support different predictions, will be considered and treated differently.



This **equalisation and differentiation**, depending on the domains in which it is used and on the purposes that it is meant to serve, may affect **positively or negatively** the individuals concerned.

Example: use of machine learning technologies to detect or anticipate health issues

- Beneficial application
- **Benefits concern in principle all data subjects** i.e., those) whose data are processed for this purpose
- Processing of health-related data may also be justified on grounds of **public health** (Article 9 (2)(h)), and in particular for the purpose of “monitoring epidemics and their spread” (Recital 46).
- This provision has become hugely relevant in the context of the Coronavirus disease 2019 (COVID-19) epidemics. In particular a vast debate has been raised by development **of applications for tracing contacts.**



Example: use of machine learning technologies to detect or anticipate health issues

➤ Such processing should be viewed as **legitimate** as long as it effectively contributes to **limit the diffusion and the harmfulness of the epidemics**, assuming that the **privacy and data protection risks are proportionate to the expected benefit**, and that **appropriate mitigation measures are applied**.

(See the European Data Protection Board Guidelines 04/2020 on the use of location data and contact-tracing tools in the context of the COVID-19 outbreak).



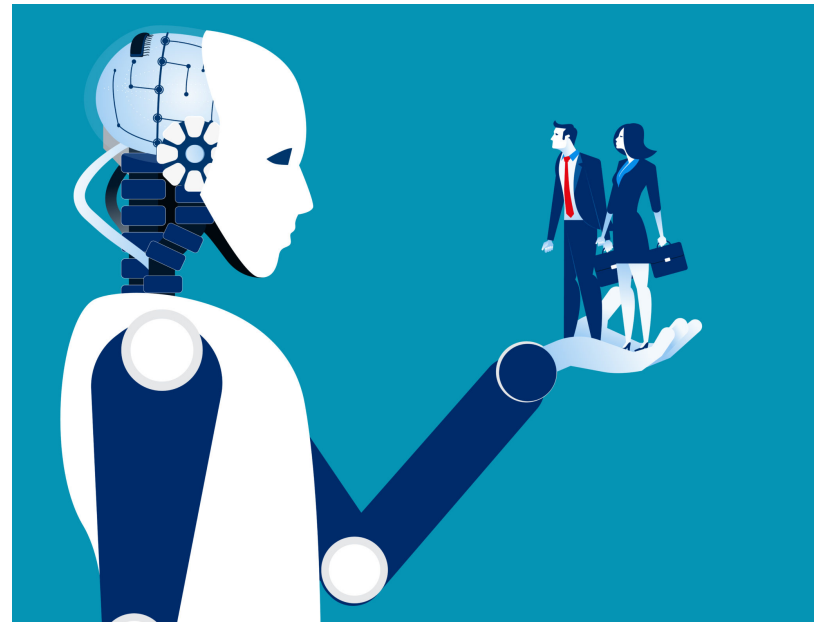
Example: use of the predictions based on health data in the contexts of insurance and recruiting

- Predictions based on health data in the context of **insurance** deserves a **much less favourable assessment**
- **Gainers:** the insured individuals getting a better deal based on their favourable health prospects.
- **Losers:** those getting a worse deal because of their unfavourable prospects.
- Individuals who already are disadvantaged because of their medical conditions would suffer further disadvantage, being excluded from insurance or being subject to less favourable conditions.



Example: use of the predictions based on health data in the contexts of insurance and recruiting

- Insurance companies having the ability to distinguish the risks concerning different applicants would have a **competitive advantage**, being able to provide better conditions to less risky applicants, so that insurers would be pressured to collect as much personal data as possible.
- **Even less commendable would be the use of health predictions in the context of recruiting**, which would involve burdening less healthy people with unemployment or with harsher work conditions. Competition between companies would also be affected, and pressure for collecting health data would grow.



A further example concerns...

Price discrimination: different prices and different conditions to different consumers, depending on predictions on their willingness to pay.

Risks:

- harm consumers
- individuals may be deprived of access to some opportunities
- affect the functioning of markets
- may be unfair(?)
- undermines the efficiency of the economy



Example: price discrimination

Conclusions..

- AI enables new kinds of algorithmic mediated differentiations between individuals
- In the **AI era differential treatments** can be based on **vast amounts of data enabling probabilistic predictions, which may trigger algorithmically predetermined responses.**
- The impacts of such practices can go beyond the individuals concerned, and affect important social institution, in the economical and political sphere.

Conclusions..

The **GDPR** provides some **constraints**:

- the need for a legal basis for any processing of personal data
- obligations concerning information and transparency
- limitations on profiling and automated decision making
- requirements on anonymisation and pseudonymisation, etc.

These constraints need to be coupled with **strong public oversight**, to ban socially obnoxious forms of differential treatment, and to adopt effective measures that prevent abuses.

Some interests at stake

- **Interest in data protection and privacy**, namely, the interest in a lawful and proportionate processing of personal data subject to oversight.
- **Interest in fair algorithmic treatment**: concern from an algorithmic transparency/explicability standpoint
- **Individual autonomy**: black box models whose functioning is not accessible and whose decisions remain unexplained and thus unchallengeable.
- **Interest in not being misled or manipulated by AI systems and to trust** such systems.
- **Indirect interest in fair algorithmic competition**, i.e., in not being subject to market-power abuses resulting from exclusive control over masses of data and technologies.

SOCIAL EMPOWERMENT



AI technologies for social and legal empowerment

Regulatory instruments and their implementation by public bodies are an essential element but they may be insufficient.

- AI & Big Data are employed in domains already characterized by a **vast power imbalance**, which they may contribute to accentuate.
- The **countervailing power of civil society is needed** to detect abuses, inform the public, activate enforcement, etc.
- In the AI era, an effective **countervailing power needs also to be supported by AI**

Citizen-empowering technologies - Claudette

Examples of citizen-empowering technologies:

ad-blocking systems

anti-spam software

anti-phishing techniques.

A step forward: services deployed with the goal of analysing and summarizing massive amounts of product reviews or comparing prices across a multitude of platforms.

One example in this direction is offered by CLAUDETTE (<https://claudette.eui.eu/>)

Citizen-empowering technologies – PDA/CDA

- Proposals for automatically extracting, categorizing and summarizing information from **privacy documents**, and assisting users in processing and understanding their contents.
- Multiple AI methods to support data protection could be merged into integrated PDA-CDA (**Privacy digital assistants/consumer digital assistants**), meant to prevent excessive/unwanted/unlawful collection of personal data and well as to protect users from manipulation and fraud, provide them with awareness of fake and untrustworthy information, and facilitate their escape from “**filter bubbles**” (**the unwanted filtering/pushing of information**).